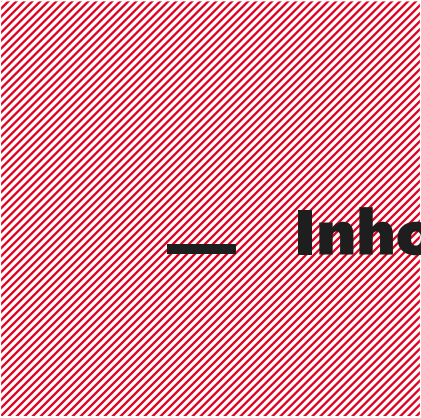


— Mobile IoT Apps 2

Angular & ASP.NET





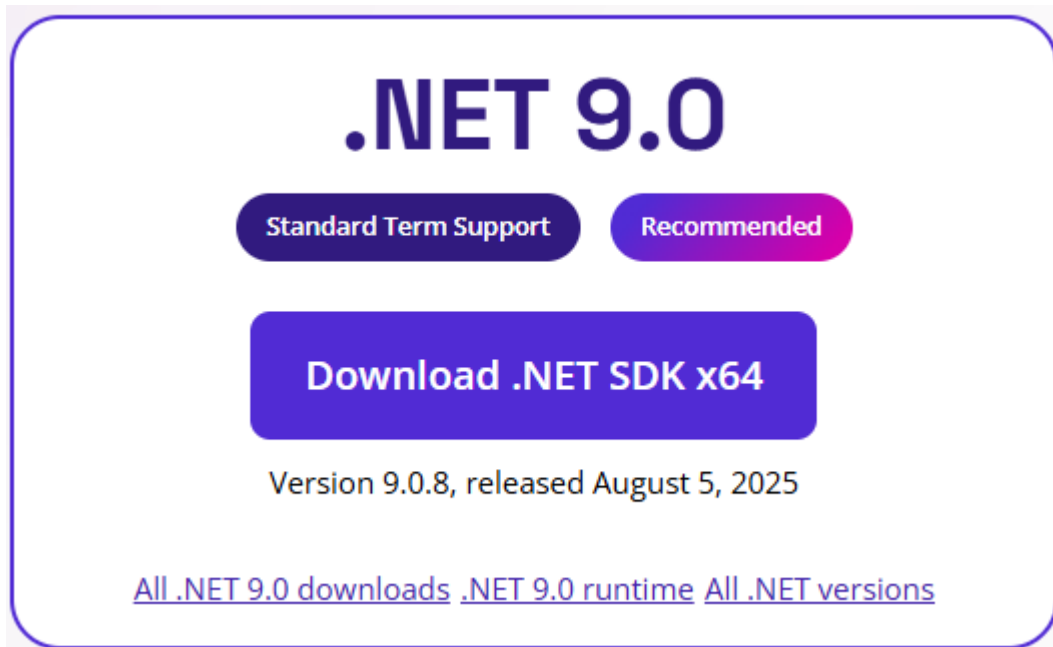
— Inhoud

- Web API
- Web API met controllers
- Web API met MongoDB
- Angular App - die gebruik maakt van Web API
- SignalR toevoegen aan Web API
- SignalR toevoegen aan Angular App

— Angular en ASP.NET

Web API

Installatie donetnet SDK [Download .NET \(Linux, macOS, and Windows\) \(microsoft.com\)](https://microsoft.com/download/.NET)



The screenshot shows the Microsoft .NET 9.0 download page. At the top, it says ".NET 9.0" in large blue font. Below this, there are two buttons: "Standard Term Support" in a dark blue rounded rectangle and "Recommended" in a purple rounded rectangle. Below these is a large blue button that says "Download .NET SDK x64". Underneath the button, it says "Version 9.0.8, released August 5, 2025". At the bottom, there are three links: "All .NET 9.0 downloads", ".NET 9.0 runtime", and "All .NET versions".

.NET 9.0

Standard Term Support Recommended

Download .NET SDK x64

Version 9.0.8, released August 5, 2025

[All .NET 9.0 downloads](#) [.NET 9.0 runtime](#) [All .NET versions](#)

— Angular en ASP.NET

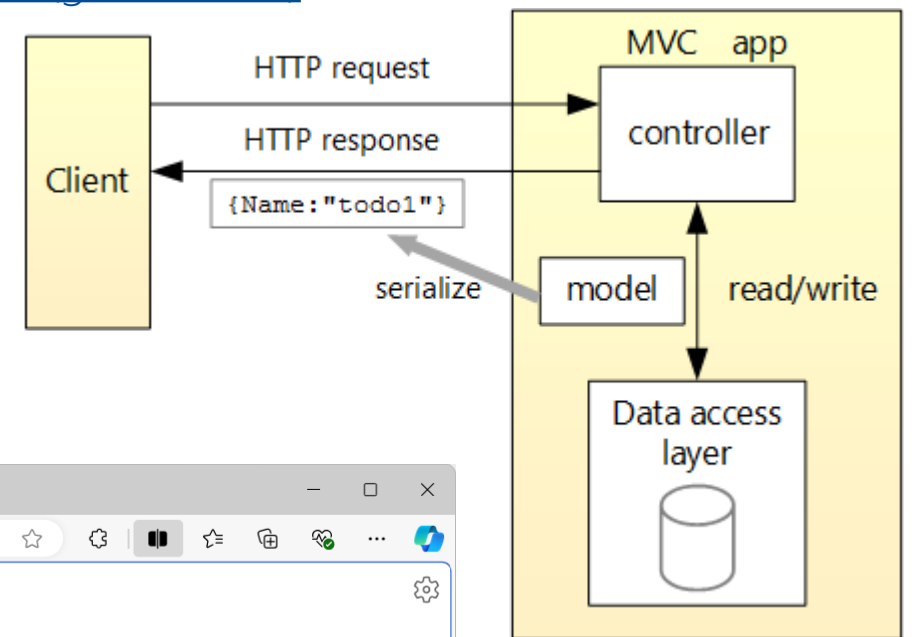
Web API met Controllers

Web API met Controllers: [Tutorial: Create a web API with ASP.NET Core | Microsoft Learn](https://learn.microsoft.com/en-us/aspnet/core/web-api/tutorial)

Resultaat: [hans-naert/ASP.NET-Core-controller-based-web-API \(github.com\)](https://github.com/hans-naert/ASP.NET-Core-controller-based-web-API)

Opmerkingen:

- Nieuw project bevat Weatherforecast controller op localhost:<portnumber>/weatherforecast
- Standaard geen swagger maar OpenAPI JSON op localhost:<portnumber>/openapi/v1.json



```
1 {
2   "openapi": "3.0.1",
3   "info": {
4     "title": "TodoApi2 | v1",
5     "version": "1.0.0"
6   },
7   "servers": [
8     {
9       "url": "https://localhost:7091"
```

```
1 [
2   {
3     "date": "2024-11-22",
4     "temperatureC": -2,
5     "temperatureF": 29,
6     "summary": "Bracing"
7   },
8   {
9     "date": "2024-11-23"
```

— Angular en ASP.NET

Web API met Controllers

Toevoegen Swagger:

- dotnet add package Swashbuckle.AspNetCore

- In Program.cs

```
builder.Services.AddEndpointsApiExplorer();  
builder.Services.AddSwaggerGen();
```

```
var app = builder.Build();
```

```
// Configure the HTTP request pipeline.
```

```
if (app.Environment.IsDevelopment())
```

```
{
```

```
    app.MapOpenApi();
```

```
    app.UseSwagger();
```

```
    app.UseSwaggerUI();
```

```
    app.UseSwaggerUI(options => // UseSwaggerUI is called only in Development.
```

```
    {
```

```
        options.SwaggerEndpoint("/swagger/v1/swagger.json", "v1");
```

```
        options.RoutePrefix = string.Empty;
```

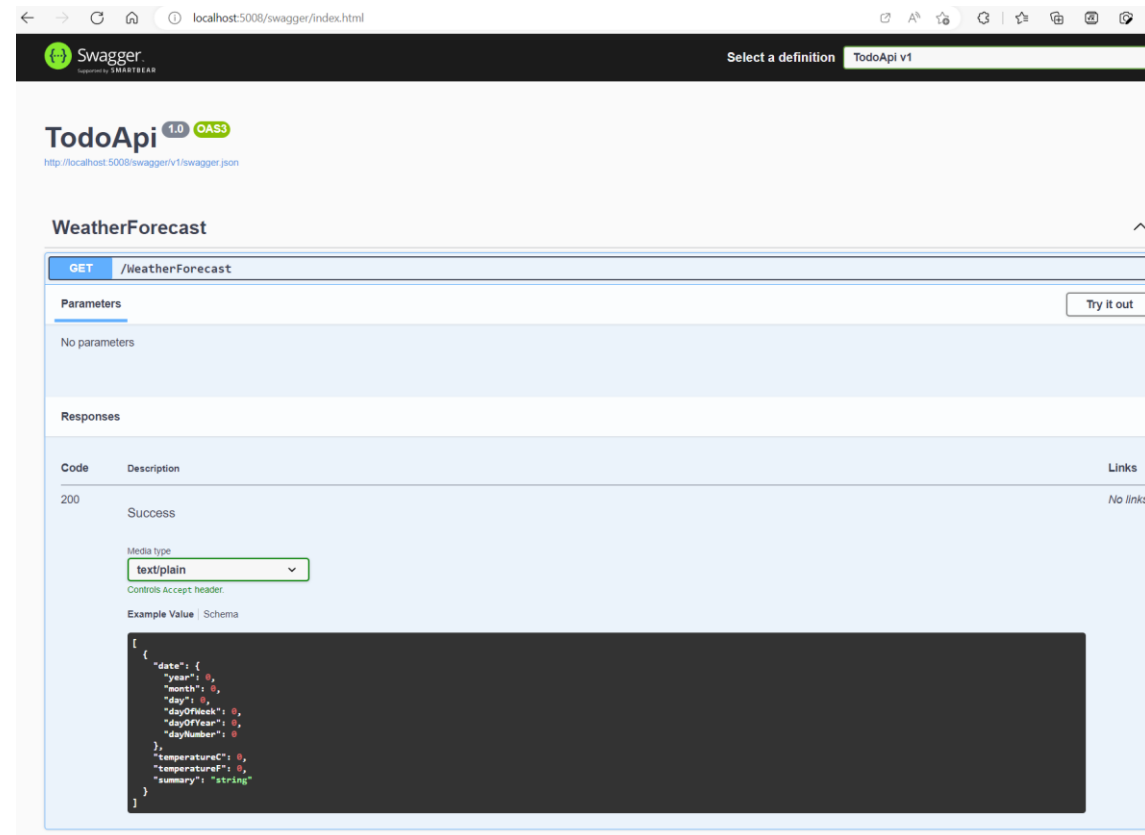
```
    });
```

```
}
```

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Web API met Controllers

- In VSCode: CTRL+F5 => RUN
- voeg /swagger toe aan URL
- Swagger: API documentation



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Web API met Controllers

- [Toevoegen model "TodoItem" in folder Models](#)

```
namespace TodoApi.Models;
```

```
public class TodoItem  
{  
    public long Id { get; set; }  
    public string? Name { get; set; }  
    public bool IsComplete { get; set; }  
}
```

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Web API met Controllers

- [database context toevoegen](#) in bestand TodoContext.cs in de Models folder

```
using Microsoft.EntityFrameworkCore;

namespace TodoApi.Models;

public class TodoContext : DbContext
{
    public TodoContext(DbContextOptions<TodoContext> options)
        : base(options)
    {
    }

    public DbSet<TodoItem> TodoItems { get; set; } = null!;
}
```


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Web API met Controllers

- [Entity Framework | Microsoft Learn](#)

Entity Framework (EF) is an object-relational mapper that enables .NET developers to work with relational data using domain-specific objects. It eliminates the need for most of the data-access code that developers usually need to write.

- [Creating a Model - EF6 | Microsoft Learn](#)

	I just want to write code...	I want to use a designer...
I am creating a new database	Use Code First to define your model in code and then generate a database.	Use Model First to define your model using boxes and lines and then generate a database.
I need to access an existing database	Use Code First to create a code based model that maps to an existing database.	Use Database First to create a boxes and lines model that maps to an existing database.

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Web API met Controllers

- [database context registreren](#) in program.cs, zodat die beschikbaar is via dependency injection in TodosContext

```
using Microsoft.EntityFrameworkCore;  
using TodoApi.Models;
```

```
var builder = WebApplication.CreateBuilder(args);
```

```
// Add services to the container.
```

```
builder.Services.AddControllers();  
builder.Services.AddDbContext<TodosContext>(opt =>  
    opt.UseInMemoryDatabase("TodoList"));
```

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Web API met Controllers

Scaffold de controller

- voeg de nuget packages toe en installeer de code-generator:

```
dotnet add package Microsoft.VisualStudio.Web.CodeGeneration.Design
dotnet add package Microsoft.EntityFrameworkCore.Design
dotnet add package Microsoft.EntityFrameworkCore.SqlServer
dotnet tool uninstall -g dotnet-aspnet-codegenerator
dotnet tool install -g dotnet-aspnet-codegenerator
```

- installeer EntityFrameworkCore.Tools:
- genereer de TodoItemsController met de code-generator

```
dotnet-aspnet-codegenerator controller -name TodoItemsController -async -api
-m TodoItem -dc TodoContext -outDir Controllers
```

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Web API met Controllers

Scaffold de controller

- PostTodoItem
- location header URI: GET /api/TodoItems/{id}

GET /api/TodoItems/{id}

Parameters

Name	Description
id * required	
integer(\$int64)	1
(path)	

Execute

Responses

Curl

```
curl -X 'GET' \
'http://localhost:5008/api/TodoItems/1' \
-H 'accept: text/plain'
```

Request URL

```
http://localhost:5008/api/TodoItems/1
```

Server response

Code	Details
200	Response body <pre>{ "id": 1, "name": "test", "isComplete": true }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Tue, 15 Nov 2022 19:24:28 GMT server: Kestrel transfer-encoding: chunked</pre>

Responses

Code	Description
200	Success Media type text/plain Controls Accept header. Example Value Schema <pre>{ "id": 0, "name": "string", "isComplete": true }</pre>

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Web API met Controllers

Routing and URL paths

```
namespace TodoApi.Controllers
{
    [Route("api/ToDoItems")]
    [ApiController]
    public class TodoItemsController : ControllerBase
```

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Web API met Controllers

- [Test web APIs with the HttpRepl | Microsoft Learn](#)
 - dotnet tool install -g Microsoft.dotnet-httprepl
 - httprepl <http://localhost:5008>
 - get /api/todoitems

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Web API met Controllers

- [Prevent over-posting](#)
 - DTO: Data Transfer Object
beperkte versie van het model - voornamelijk voor security, bvb secret veld niet versturen

- voeg veld Secret toe aan TodoItem.cs

```
public string? Secret { get; set; }
```

- maak DTO:

```
namespace TodoApi.Models;
```

```
public class TodoItemDTO
{
    public long Id { get; set; }
    public string? Name { get; set; }
    public bool IsComplete { get; set; }
}
```

- Maak gebruik van functie ItemToDTO

```
private static TodoItemDTO ItemToDTO(TodoItem todoItem) => new TodoItemDTO { Id = todoItem.Id,
Name = todoItem.Name, IsComplete = todoItem.IsComplete };
```

— Angular en ASP.NET

Web API en MongoDB

- [Create a web API with ASP.NET Core and MongoDB | Microsoft Learn](#)
Resultaat: [hans-naert/ASP.NET-Core-web-API-MongoDB \(github.com\)](https://github.com/hans-naert/ASP.NET-Core-web-API-MongoDB)
 - MongoDB=NoSQL database
 - MongoDB is a document database with the scalability and flexibility that you want with the querying and indexing that you need
bron [What Is MongoDB? | MongoDB](#)

connectiestring te plaatsen in appsettings.json en
appsettings.Development.json

"ConnectionString":

"mongodb+srv://webapi:WZD8LORzFGLTay6C@cluster-2023-2024.j5igczs.mongodb.net/",

```
1  {
2    _id: "5cf0029caff5056591b0ce7d",
3    firstname: 'Jane',
4    lastname: 'Wu',
5    address: {
6      street: '1 Circle Rd',
7      city: 'Los Angeles',
8      state: 'CA',
9      zip: '90404'
10   }
11 }
```



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Web API en MongoDB

- Installeer
 - MongoDB
 - Op laptop [MongoDB Community Server | MongoDB](#): winget install MongoDB.Server -i
 - Cloud: [MongoDB Atlas | Multi-cloud Developer Data Platform | MongoDB](#)
 - MongoDB Shell [Try MongoDB Tools - Download Free Here | MongoDB](#)
winget install MongoDB.Shell -i
- [Configuratie van MongoDB](#)
 - PATH updaten
 - met mongod: C:\Program Files\MongoDB\Server\<version_number>\bin
 - met mongosh: C:\Software\mongosh-1.6.0-win32-x64\bin
 - mongod --dbpath <data_directory_path>

— Angular en ASP.NET

Web API en MongoDB

- mongosh <connectiestring>
 - use BookStore
 - db.createCollection('Books')
 - db.Books.insertMany([{ "Name": "Design Patterns", "Price": 54.93, "Category": "Computers", "Author": "Ralph Johnson" }, { "Name": "Clean Code", "Price": 43.15, "Category": "Computers", "Author": "Robert C. Martin" }])
 - db.Books.find().pretty()

```
BookStore> db.Books.find().pretty()
[
  {
    _id: ObjectId('6747489ef828699b950d8190'),
    Name: 'Design Patterns',
    Price: 54.93,
    Category: 'Computers',
    Author: 'Ralph Johnson'
  },
  {
    _id: ObjectId('6747489ef828699b950d8191'),
    Name: 'Clean Code',
    Price: 43.15,
    Category: 'Computers',
    Author: 'Robert C. Martin'
  }
]
```

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Web API en MongoDB

- [Create the ASP.NET Core web API project](#)
 - dotnet new webapi -o BookStoreApi
 - code BookStoreApi
 - dotnet add package MongoDB.Driver
- [Add an entity model](#) Models/Book.cs:

```
using MongoDB.Bson;
using MongoDB.Bson.Serialization.Attributes;

namespace BookStoreApi.Models;
public class Book
{
    [BsonId]
    [BsonRepresentation(BsonType.ObjectId)]
    public string? Id { get; set; }
    [BsonElement("Name")]
    public string BookName { get; set; } = null!;
    public decimal Price { get; set; }
    public string Category { get; set; } = null!;
    public string Author { get; set; } = null!;
}
```

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Web API en MongoDB

- [Add a configuration model](#)
 - aanpassen appsettings.json
 - BookStoreDataBaseSettings class toevoegen onder Models
 - in Program.cs - settings van appsettings.json verkrijgen door:
- [Add a CRUD operations service](#)
 - BooksService in Services folder toevoegen met CRUD opdrachten
 - BookService als Singleton toevoegen aan Program.cs
- [Add a controller](#)
 - Voeg BooksController toe aan de Controllers folder
deze gebruikt de BooksService class om de CRUD opdrachten uit te voeren

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Web API en MongoDB

- [Test the web API](#)
 - dotnet run

← → ↻ 🏠 ⓘ localhost:5153/api/books

```
[{"id":"6374061b25d4a87ef087ba5a","bookName":"Design Patterns","code","price":43.15,"category":"Computers","author":"Robert C. M
```

← → ↻ 🏠 ⓘ localhost:5153/api/books/6374061b25d4a87ef087ba5a

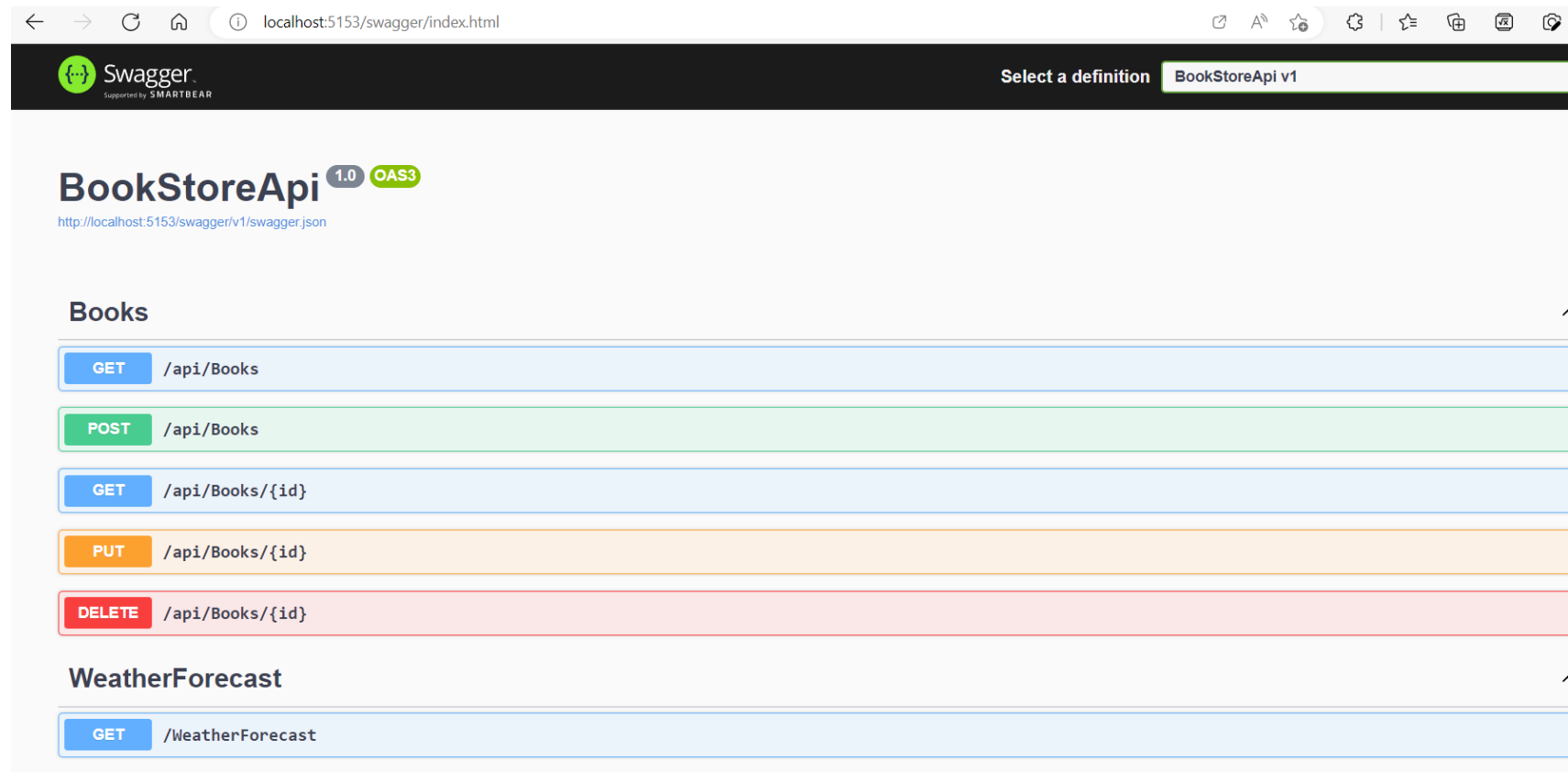
```
{"id":"6374061b25d4a87ef087ba5a","bookName":"Design Patterns","price":54.93,"cate
```

- [Configure JSON serialization options](#)
 - Wat:
 - naamgeving camel casing => Pascal Casing (bvb bookName => BookName)
 - de naam van een property wijzigen t.o.v. naamgeving in de code
 - Hoe:
 - Program.cs: AddJsonOptions aan AddControllers toevoegen
 - attribuut [JsonPropertyName("Name")]

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Web API en MongoDB

- Swagger



— Angular en ASP.NET

Angular App - die gebruik maakt van Web API

- Maak een Angular App die communiceert met de Web API met MongoDB database die:
 - een lijst van boeken toont
 - een form weergeeft waarmee je een boek kunt toevoegen

Voorbeeld oplossing:

[hans-naert/angular-webapi \(github.com\)](https://github.com/hans-naert/angular-webapi)

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Angular App - die gebruik maakt van Web API

- Maak een lege angular App:
 - indient geen ntfs:
git config --global --add safe.directory D:/HN/Mobile_IoT_Apps_2/voorbereiding/angular-webapi
 - ng new angular-webapi
 - cd angular-webapi
 - ng add @angular/material
 - ng g component fetch-data
=> plaats daar een fetch in naar <localhost:5008/weatherforecast>
=> code van fetch-data component gebaseerd op my-new-app
 - CORS fout:

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Angular App - die gebruik maakt van Web API

- Als de response van een ander domein geen CORS headers meegeeft => CORS fout

```
Angular is running in development mode. Call enableProdMode() to enable production mode.
✖ Access to XMLHttpRequest at 'http://localhost:5008/weatherforecast' from origin 'http://localhost:4200' has been blocked by CORS policy: No 'Access-Control-Allow-Origin' header is present on the requested resource.
  ▶
  ▶ HttpErrorResponse {headers: HttpHeaders, status: 0, statusText: 'Unknown Error', url: 'http://localhost:5008/weatherforecast', o
    k: false, ...}
  ▶ GET http://localhost:5008/weatherforecast net::ERR_FAILED 200
```

- Waarom CORS => [What is CORS \(cross-origin resource sharing\)? Tutorial & Examples | Web Security Academy](#)
- CORS headers toevoegen aan Web API: [Enable Cross-Origin Requests \(CORS\) in ASP.NET Core | Microsoft Learn](#)

```
builder.Services.AddCors(
    o => o.AddPolicy("CorsPolicy", builder => { builder
        .AllowAnyMethod()
        .AllowAnyHeader()
        .AllowCredentials()
        .WithOrigins("http://localhost:4200"); }) );
app.UseCors("CorsPolicy");
```

— Intermezzo

NuGet.config

- problemen bij ophalen packages => controleren of NuGet.config correct is
C:\Users\username\AppData\Roaming\NuGet

⚙ NuGet.Config ✕

C: > Users > hansn > AppData > Local > Microsoft > Windows > INetCache > Content.Outlook > 4F92H90A > ⚙ NuGet.Config

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <configuration>
3   <packageSources>
4     <add key="nuget.org" value="https://api.nuget.org/v3/index.json" protocolVersion="3" />
5   </packageSources>
6 </configuration>
```

— Intermezzo

global.json

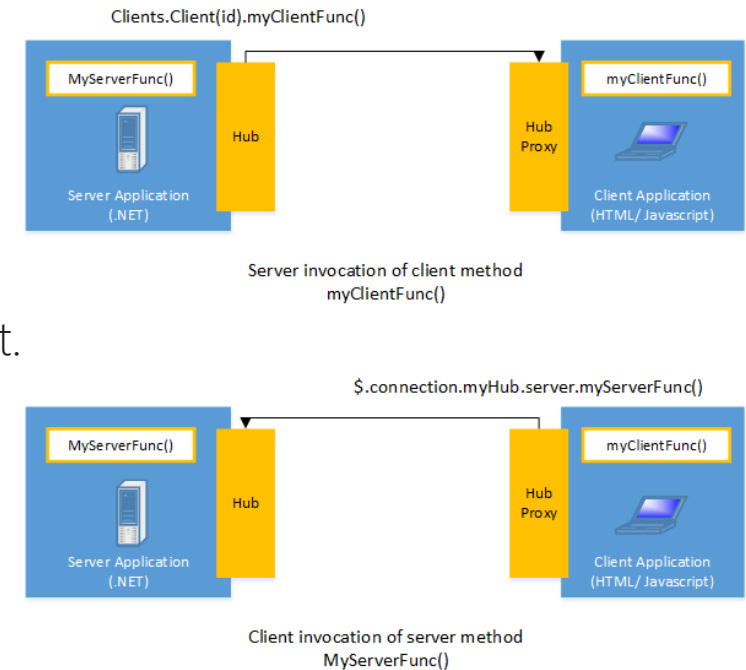
- Selectie van .NET SDK => via global.json wordt gezocht in werkdirectory en parent folders
[global.json overview - .NET CLI | Microsoft Learn](#)
- inhoud global.json:

```
{  
  "sdk": {  
    "version": "8.0.100" }  
}
```

— Angular en ASP.NET

SignalR toevoegen aan Web API

- SignalR: [Introduction to SignalR | Microsoft Learn](#)
SignalR biedt de mogelijkheid om berichten te pushen naar de client. Dit wordt achter de schermen gedaan met WebSockets.
- SignalR in ASP.NET <-> Socket.io in node.js
- Uitvoeren van de Tutorials
 - [Get started with ASP.NET Core SignalR | Microsoft Learn](#)
 - [Tutorial: Get started with ASP.NET Core SignalR using TypeScript and Webpack | Microsoft Learn](#)



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SignalR toevoegen aan Angular App

Voorbeeld van Angular App met backend die SignalR gebruikt:

- backend: SignalRChatHub: [hans-naert/SignalRChat \(github.com\)](https://github.com/hans-naert/SignalRChat)
- angular app: [hans-naert/angular-signalr \(github.com\)](https://github.com/hans-naert/angular-signalr)

Oefening

- Voeg SignalR toe aan de Web API (aanpassen [hans-naert/ASP.NET-Core-web-API-MongoDB \(github.com\)](https://github.com/hans-naert/ASP.NET-Core-web-API-MongoDB))
- Voeg SignalR toe aan de Angular App (aanpassen [hans-naert/angular-webapi \(github.com\)](https://github.com/hans-naert/angular-webapi))
- Toon een lijst met de TODO's, via SignalR laat je de lijst updaten als er een item wordt toegevoegd